

MARKER SELECTION

Learned gene-gene graph



Expression



$$x \in \mathbb{R}^N$$

(N genes)

smooth x (denoise) $\hat{G}_t$ — low-rank, end-to-end

Smoothed expression

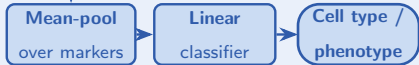
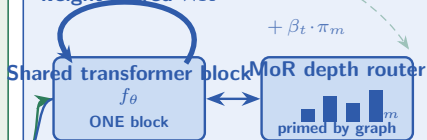
$$\tilde{x} = \hat{G} x$$

Q-Former marker router



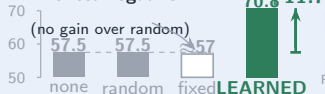
RECURSIVE ROUTING & CLASSIFICATION

weight-shared $\times K$



Gene-graph ablation — single-cell macro-F1

honest negative



4x fewer
params @ K=4

~38% fewer
FLOPs, matched acc.